EASTERN MEDITERRANEAN UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG345
Course Name: Heat Transfer
Semester/ Year: Spring 2024-2025
Instructor: Assoc. Prof. Dr. Murat Özdenefe
Assistant: Mr. Hussain Shawish
Assessment: Lab
Due Date: 18 March 2025 at 23:59 hrs.

LAB NAME: Steady State Conduction Through a Uniform Wall

STUDENT NO.: 23704019 **NAME, SURNAME: TARIK TARIQ**

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
S-1	6	1, 3	10		
S-2	6	1, 3	10		
S-3	6	1, 3	10		
S-4	6	1, 3	10		
S-5	6	1, 3	30		
S-6	6	1, 3	30		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

	Course Learning Outcomes	Student Outcomes 1	2	3	4	5	6	7
1	Understand the basic mechanisms of heat transfer.	X					X	
2	Be able to obtain and solve DE of heat conduction.	X						
3	Comprehend steady conduction, solve steady conduction problems by resistance concept and analyze fins.	X					X	
4	Be able to solve transient heat transfer problems for lumped systems & systems with spatial effects.	X						
5	Understand the numerical solution of steady, 1D & 2D and transient, 1D & 2D heat conduction problems.	X						
6	Evaluate convection coefficient, heat transfer and associated temperatures for external flow.	X					X	
7	Calculate convection coefficient, heat transfer and associated temperatures for internal flow.	X						
8	Evaluate convection coefficient and heat transfer for free convection cases.	X						
9	Analyze & size HEXs by LMTD and ε-NTU method.	X						
	Weight of Student Outcomes	H					M	

Instructions:

A. Report presentation: Draw neat, labeled diagrams where necessary. Write relevant equations and write your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.

B. Submission: Upload your report as a pdf file to the “Lab Work and Project” section in LMS within the designated time. The file uploaded should be named as [MENG345-Lab1-Student ID-Student Name]. No late submissions will be accommodated.

ABSTRACT

This experiment investigates steady-state heat conduction through a uniform solid wall using the HT10XC Heat Transfer Service Unit. The primary objective was to verify Fourier's law by analyzing the temperature distribution and heat transfer rate under different voltage settings. Temperature measurements were recorded at various points along the heated and cooled sections, allowing for the determination of thermal gradients and heat flux. Results showed that as the heat transfer rate increased, the temperature differences ΔT across both the heated and cooled regions also increased, demonstrating a direct proportionality between heat flux and temperature gradient. The cooling region exhibited consistently higher temperature differences compared to the heated section, suggesting variations in thermal resistance or cooling efficiency. The calculated values of the constant C showed minor variations across iterations, likely due to experimental uncertainties such as sensor precision, heat losses, or thermal contact resistance. Overall, the findings align with theoretical predictions, confirming Fourier's law of heat conduction. The experiment highlights the importance of understanding steady-state conduction for applications in thermal insulation, electronic cooling, and structural heat management.

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1. INTRODUCTION

Heat transfer is a fundamental concept in engineering that governs the movement of thermal energy due to temperature differences. One of the primary modes of heat transfer is conduction, which occurs within solid materials. Steady-state conduction refers to the condition in which the temperature distribution within the material remains constant over time, meaning that the heat entering the system is equal to the heat leaving it. Understanding this phenomenon is crucial for applications in thermal insulation, electronic cooling, and structural heat management [1].

The study of heat conduction dates back to the early work of Joseph Fourier in the 19th century. Fourier's law of heat conduction establishes that the rate of heat transfer through a material is proportional to the negative temperature gradient and the material's thermal conductivity. This principle forms the foundation for analyzing steady-state conduction in engineering applications [2].

In this experiment, the **HT10XC Heat Transfer Service Unit** was used to investigate steady-state conduction through a uniform wall. The system allowed for the measurement of temperature at various points along the heated and cooled regions of the wall. Specifically, **T1, T2, and T3** were measured in the heated region, while **T6, T7, and T8** were recorded in the cooling region. These measurements facilitated the determination of temperature gradients and heat flux across the material, which can be analyzed using Fourier's law [3].

2. THEORY

Heat conduction is a mode of heat transfer that occurs within a solid material due to a temperature gradient. In steady-state conduction, the temperature at any given point in the material remains constant over time, meaning the heat entering the system is equal to the heat leaving it. This principle is governed by Fourier's law, which states that the rate of heat transfer through a solid is directly proportional to the temperature difference and the cross-sectional area while being inversely proportional to the thickness of the material[3].

Mathematically, Fourier's law for one-dimensional heat conduction through a plane wall can be expressed as:

$$\dot{Q} = -kA\, dT/dx$$

where:

- $\dot{Q}$ is the rate of heat transfer (W),
- **k** is the thermal conductivity of the material (W/m·K),

- **A** is the cross-sectional area perpendicular to heat flow (m^2),
- **dT/dx** is the temperature gradient across the material (K/m).

In this experiment, the heated and cooled sections are tightly clamped together, ensuring good thermal contact between the faces. This allows the system to be considered as a single, continuous wall with uniform material properties. By measuring the temperature distribution and calculating the heat transfer rate, the validity of Fourier's law can be examined by the following relationship.

$$\dot{Q} \propto \Delta T/\Delta x$$

K (thermal conductivity) and **A (area)** are not changing throughout the experiment and can both be treated as constants **C**.

$$C = KA$$

3. APPARATUS

The **HT11C Linear Heat Conduction Accessory**, used with the **HT10XC Heat Transfer Service Unit**, studies one-dimensional heat conduction by measuring temperature distribution along a solid conductor. It includes a heated and cooled section, with sensors recording temperature gradients to analyze thermal conductivity and verify Fourier's Law[4].

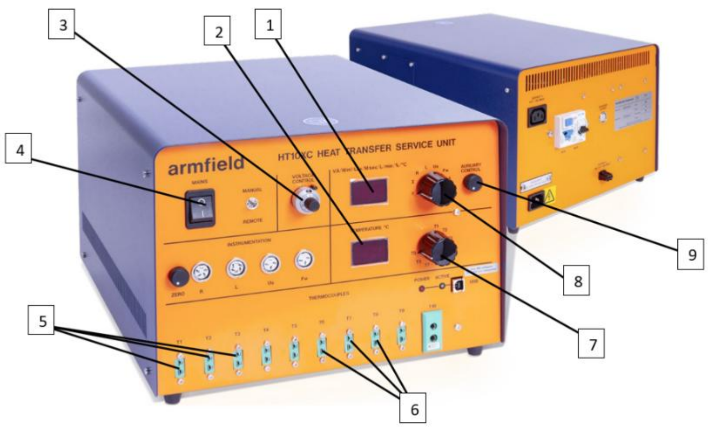


Figure 3.1 HT10XC heat transfer service unit

The **HT10XC Heat Transfer Service Unit**:

1. **Digital Display (Voltage & Current)** – Shows the voltage (V) and current (A) applied to the system.
2. **Digital Display (temperature)** – Displays temperature across the system.
3. **Voltage Control Knob** – Adjusts the voltage supplied to the heating system to control the temperature.
4. **Main Power Switch (Mains)** – Turns the unit on or off; enables power supply to the system.
5. **Thermocouple Input Ports** – Connects temperature sensors (T1, T2, T3) for measuring temperature at different points in the experiment.
6. **Thermocouple Input Ports** – Connects temperature sensors (T6, T7, T8) for measuring temperature at different points in the experiment.
7. **Temperature Knob** – Used for selecting temperature measurement points and controlling various settings.
8. **Temperature Knob** – Used for selecting temperature measurement points and controlling various settings.
9. **Auxiliary Control Knobs** – Used Adjusts the voltage supplied to the heating system to control the temperature.

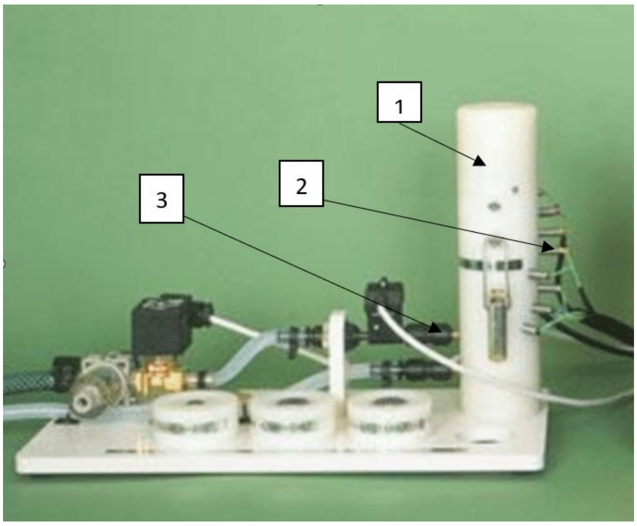


Figure 3.2 Computer Controlled Linear Heat Conduction

1. **White Cylinder**

- Contains the heating element or test section.
- Inner core contains a cylindrical conductor.
- Contains an insulating white outer layer to prevent heat transfer in the radial direction.
- Used to apply controlled heat for experiments.

- Equipped with temperature sensors for monitoring.

2. **Thermocouples temperature sensors.**

- Upper sensors used to measure the temperatures across the heated region
- Lower sensors used to measure temperature across the cooling region.

3. **Cooling system**

- Regulates fluid flow (water) to the test section
- Includes solenoid valves, flow meters, and pressure regulators.

4. PROCEDURE

Start by checking if the panel displays on the service console are turned on. If they are not, check the RCD switch at the back of the service unit and make sure it is in the "up" position. If you are using a computer, make sure the software shows "IFD OK" in the bottom right corner of the screen.

Next, turn on the cooling water and set the flow control valve (not the pressure regulator) to about 1.5 liters per minute. If using the software, adjust and monitor the flow rate using the control box in the software's mimic diagram window. If not using the software, use the selector switch (25) on the console to show the flow rate on the display in liters per minute and adjust it using the AUXILIARY CONTROL knob. If using the HT11 model, the flow rate is controlled manually with a valve next to the test section.

Once the flow rate is set, adjust the heater voltage to 9 volts. If using the computer, enter the voltage in the heater display box or use the control arrows. If using the console, adjust the voltage control potentiometer until the top panel shows a reading of 9 volts with the selector switch set to position V.

Wait for the HT11C to stabilize. If using a computer, monitor the temperatures on the software screen. If using the console, use the lower selector switch (28) to display the temperature readings from each sensor. Once the temperatures are stable, record the values for T1, T2, T3, T6, T7, T8, V I, and Fw. If using the console, write down these values manually. The Fw reading is not available unless the optional SFT2 flow sensor is installed.

Then, set the heater voltage to 9 volts and allow the system to stabilize. Repeat the process and record the same temperature readings. After that, increase the heater voltage to 12 volts, wait for the system to stabilize, and take the same readings. Finally, set the heater voltage to 17 volts, allow the system to stabilize, and record the values once again.

5. MEASUREMENTS, CALCULATIONS AND RESULTS

The collected data was organized into a table showing the temperature values for each iteration, and the results were further analyzed by plotting graphs to illustrate the temperature variations at different voltage levels. These calculations and graphical representations provide a clear understanding of how the voltage settings impact the heat transfer performance in both regions.

Table 5.1 Experimental Data

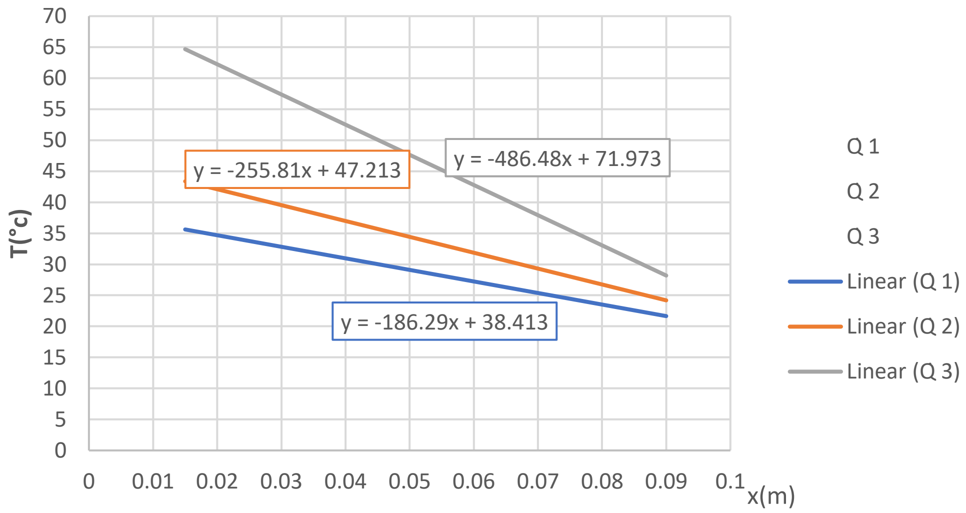
	X(m)	T(°c) 1st iteration	T(°c) 2nd iteration	T(°c) 3rd iteration
1	0.15	35.1	42.7	69.8
2	0.03	32.8	39.3	57.5
3	0.045	29.8	35.3	50.5
6	0.06	29	34.5	44
7	0.075	24.3	28.3	35.7
8	0.09	20.8	22.6	27.1

Table 5.2 shows the different voltage readings and current readings. For the first iteration, we started with 9V and repeated the experiment by changing the voltage all the way to 17V.

Table 5.2 Current & Voltage table

Iteration	Current (I)	Voltage (V)
1	0.99	9
2	1.32	12
3	1.86	17

Graph 5.1 shows the relationship between temperature and the distances of each thermocouple or temperature sensors. A line of best fit was constructed using excel in order to obtain the gradient of each line. All three lines have negative gradients which means there was an inverse relation between temperature readings and its corresponding distance.



graph 5.1 Temperature against Distance graph

Table 5.3 Change in Temperature and rate of Heat Transfer Data

	$\dot{Q}$ (W)	**Δ T_{hot} (°c)**	**Δ T_{cold} (°c)**
1	8.91	5.3	8.2
2	15.84	7.4	11.9
3	31.62	13.3	16.9

Calculations for the first iteration:

Δ T_{hot} (°c) = $T_1 - T_3$ = 35.1-29.8 **= 5.3°c**

Δ T_{cold} (°c) = $T_6 - T_8$ = 29-20.8 **= 8.2°c**

Heat transfer rate:

$\dot{Q}$=VI = 9 x 0.99= **8.91W**

Constant C:

C_1 = $\dot{Q}$/slope, slope = Δy/ Δx = ΔT/ Δx = 8.91W/-186.29 =| -0.04783| = **0.04783**

Note that the results above are for the first iteration only and the same calculations were done on the second and third iteration and the absolute value of C was taken since K & A are positive constants.

Table 5.4 Table for the constant C for each iteration

iteration	Constant C = KA
1	0.04783
2	0.06192
3	0.064998

6. DISCUSSION AND CONCLUSION

The experiment demonstrated steady-state heat conduction and verified Fourier's law by analyzing temperature gradients and heat flux through a uniform wall. As the heat transfer rate increased, the temperature difference across the material also increased, confirming the expected direct proportionality. The cooling region consistently exhibited a larger temperature difference compared to the heated region, suggesting differences in thermal resistance or heat dissipation efficiency. Despite minor experimental variations, the results aligned with theoretical predictions.

1) **Comparison of** ΔT_{hot} **and** ΔT_{cold} **at the Same Heat Transfer Rate**

At each heat transfer rate, the temperature difference in the cooling region ΔT_{cold} was greater than that in the heated region ΔT_{hot}. This indicates that heat dissipation in the cooling section might be less efficient or that the material properties influence the heat distribution differently in each region.

2) Comparison of ΔT_{hot} **at Different** $\dot{Q}$

As the heat transfer rate increased, ΔT_{hot} also increases. This confirms that a higher energy input leads to a steeper temperature gradient in the heated region. The trend was mostly linear, though minor deviations may be attributed to thermal contact resistance or heat losses.

3) Comparison of ΔT_{cold} at Different $\dot{Q}$

Similar to ΔT_{hot}, ΔT_{cold} increased as the heat transfer rate increased. This supports the fundamental heat conduction principle that a greater heat flux results in a larger temperature gradient. However, the cooling region showed a consistently larger ΔT, which may be due to differences in convective cooling efficiency or material properties.

4) Comment on the Values of C

The calculated values of C varied slightly across different iterations. Ideally, this value should remain constant if the material and cross-sectional area do not change. However, small fluctuations were observed, which could be attributed to experimental uncertainties such as sensor accuracy, heat losses, or variations in thermal contact resistance. Despite these variations, the overall trend confirmed the relationship between heat transfer rate and temperature gradient, validating Fourier's law.

7. REFERENCES

[1] *Theodore L. Bergman, Adrienne S. Lavine, Frank P. Incropera, David P. De Witt Fundamentals Of Heat And Mass Transfer, Seventh Edition John Wiley & Sons, Inc. (2011) : Free Download, Borrow, and Streaming : Internet Archive*.

[2] D. Pineda-Mora *et al.*, "Emerging Water Quality Issues along Rio de la Sabana, Mexico," *J Water Resour Prot*, vol. 10, pp. 621–636, 2018, doi: 10.4236/jwarp.2018.107035.

[3] "Heat and Mass Transfer: Fundamentals and Applications," https://www.mheducation.com/highered/product/Heat-and-Mass-Transfer-Fundamentals-and-Applications-Cengel.html.

[4] "Heat Transfer Service Unit Instruction Manual," 2012.